

LINEAR ALGEBRA NOTES

DR. JISHU DAS

ABSTRACT. This is a compilation of Linear Algebra notes for the course OEC-LA taught in the academic year 2025-26 to SY B.Tech. students at COEP Technological University, Pune.

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1. VECTOR SPACES

1.1. Introduction.

Definition 1 (Vector Space). A vector space V over a field F , consists of a set on which two operations are defined so that for every pair of elements x, y , there is a unique element $x + y \in V$, and for each element $a \in F$ and $x \in V$ there is a unique element $ax \in V$, such that the following holds.

- (VS1) For all $x, y \in V$, $x + y = y + x$.
- (VS2) For all $x, y, z \in V$, $(x + y) + z = x + (y + z)$.
- (VS3) There exists an element $0_V \in V$ such that $x + 0_V = x$.
- (VS4) For all $x \in V$, there exists $y \in V$ such that $x + y = 0_V$.
- (VS5) For every $x \in V$, $1x = x$.
- (VS6) For every $a, b \in F$, and $x \in V$, $(ab)x = a(bx)$.
- (VS7) For every $a, b \in F$, and $x \in V$, $(a + b)x = ax + bx$.
- (VS8) For every $a \in F$, and $x, y \in V$, $a(x + y) = ax + ay$.

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The elements $x + y$ and ax are called the sum of x and y and the product of a and x , respectively. The elements of F are called scalars, and the elements of vectors are called vectors. Throughout the notes, 0_V denotes the zero vector of the vector space V . If a set fails to satisfy any of the conditions mentioned above, then the set is not called a vector space. **Unless specified, we will take $F = \mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers.**

Example 1. (1) Let $V_1 = \mathbb{R}^2$ and $F = \mathbb{R}$. Let $x = (x_1, x_2), y = (y_1, y_2)$, then $x + y = (x_1 + x_2, y_1 + y_2)$, and for $k \in \mathbb{R}$, $kx = (kx_1, kx_2)$.

(2) Let $V_2 = \mathbb{R}^n$ and $F = \mathbb{R}$ for $n \in \mathbb{N}$. Let $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$, then

$$x + y = (x_1 + y_1, \dots, x_n + y_n),$$

and for $k \in \mathbb{R}$, $kx = (kx_1, \dots, kx_n)$. In general, for a given field F , one can take $V_2 = F^n$ with $n \in \mathbb{N}$.

(3) Let V_3 be the set of all $m \times n$ with entries from $\mathbb{R}$. That is an element of V looks like

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix},$$

where each entry $a_{ij} \in \mathbb{R}$ for all $i = 1, \dots, m$ and $j = 1, \dots, n$. Then, V_3 is a vector space equipped with matrix addition and scalar multiplication. For convenience, we use the notation $M_{m \times n}(\mathbb{R})$ for the above vector space.

(4) Let $V_4 = \{f \mid f \text{ is a function from } \mathbb{R} \text{ to } \mathbb{R}\}$. We define $(f + g)(x) := f(x) + g(x)$ and for $k \in \mathbb{R}$, $(kf)(x) := kf(x)$. One can check that V is a vector space over F .

(5) Let V_5 be the set of polynomials with real coefficients and at most degree 3 with $F = \mathbb{R}$. That is $V = \{a + bx + cx^2 + dx^3 \mid a, b, c, d \in \mathbb{R}\}$.

Non-example 1. (1) Let $S_1 = \{x \in \mathbb{R} : x > 0\}$, $F = \mathbb{R}$. The set S is not a vector space as $1 - 1 = 0 \notin S$.

(2) Let $S_2 = \mathbb{R}^2, k \in \mathbb{R}$. For $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$, let $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 - y_2)$ and $k(x_1, x_2) = (kx_1, kx_2)$. It does not satisfy (VS1) as $(1, 2) + (3, 3) \neq (3, 3) + (1, 2)$.

(3) Let $S_3 = \mathbb{R}^2, k \in \mathbb{R}$. For $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$, let $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 y_2)$ and $k(x_1, x_2) = (kx_1, x_2)$. It does not satisfy (VS4), for instance take $x = (1, 0)$.

(4) Let $S_4 = \mathbb{R}^2, k \in \mathbb{R}$. For $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$, let $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ and $k(x_1, x_2) = (kx_1, 0)$. It does not satisfy (VS5) as $1(2, 3) = (2, 0) \neq (2, 3)$.

(5) Let S_5 be the set of polynomials with real coefficients of degree equal to 3 with $F = \mathbb{R}$. That is $S_5 = \{a + bx + cx^2 + dx^3 \mid a, b, c, d \in \mathbb{R}, d \neq 0\}$. It is not a vector space as $(-x^3 + x) + x^3 = x \notin S$.

1.2. Subspaces.

Definition 2 (Subspace). A subset W of a vector space V over F is called a subspace of V if W is a vector space with respect to the addition and scalar multiplication defined on V .

In order to check whether a subset is a subspace of a vector space V or not, we only need to see if (VS3), (VS4) hold and $x + y \in W, kx \in W$ for all $x, y \in W, k \in F$; as other axioms automatically get satisfied. The following theorem gives a criterion to check if a set W is a subspace of a vector space V or not.

Theorem 1 (Subspace criterion). *Let V be a vector space over a field F and W be a subset of V . Then W is a subspace of V if and only if the following three conditions hold for the operations defined in V .*

- (1) $0_V \in W$.
- (2) $x + y \in W$, for all $x, y \in W$.
- (3) $kx \in W$, for all $x \in W, k \in F$.

Example 2. (1) *Let V be a vector space over F . Then the set V and $\{0_V\}$ are subspaces of V . These are called trivial vector spaces for V .*

- (2) *Let $V = \mathbb{R}^3$ and $F = \mathbb{R}$. Let $W_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = x_2 = x_3\}$, then W_1 is a subspace of $\mathbb{R}^3$. Let $W_2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$, then W_2 is a subspace of $\mathbb{R}^3$.*
- (3) *Let $n \in \mathbb{N}$, and recall that $M_{n \times n}(\mathbb{R})$ denotes the set of all real square matrices of order n . Let A^T denote the transpose of a matrix $A \in M_{n \times n}(\mathbb{R})$. Let $W_3 = \{A \in M_{n \times n}(\mathbb{R}) : A^T = A\}$ i.e. the set of all real symmetric matrices of order n . Then W_3 is a subspace of V . Similarly the set of all real skew symmetric matrices of order n is a subspace of $M_{n \times n}(\mathbb{R})$.*
- (4) *Let $W_4 = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is differentiable on } \mathbb{R}\}$. Then W_4 is a subspace of V_5 in Example 1.*

Non-example 2. (1) *Let $V = \mathbb{R}^3$ and $F = \mathbb{R}$. Let $S_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = x_2 = x_3 + 1\}$, then S_1 is not subspace of $\mathbb{R}^3$ as $(0, 0, 0) \notin \mathbb{R}^3$.*

- (2) *Let $S_2 = \{A \in M_{n \times n}(\mathbb{R}) : \det(A) \geq 0\}$, then S_2 is not a subspace as $-I_n \notin M_{n \times n}(\mathbb{R})$, where I_n is the identity matrix of order n .*
- (3) *Let $W_3 = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is not differentiable on } \mathbb{R}\}$. Then S_3 is not a subspace of V_5 in Example 1 as $|x|, -|x| + 1 \in S_3$, but $|x| + (-|x| + 1) \notin S_3$.*

Proposition 1.1. *Any intersection of subspaces of a vector space V is a subspace of V .*

Proof. Let $\mathcal{A}$ be a collection of vector spaces and $\tilde{V} = \cap_{V \in \mathcal{A}} V$. Since each V in the collection contains the zero vector, $\cap_{V \in \mathcal{A}} V$ also contains the zero vector. Let $x, y \in \tilde{V}$, then $x, y \in V$ for all $V \in \mathcal{A}$. Since each $V \in \mathcal{A}$ is a vector space, $x + y$ lies in V for each $V \in \mathcal{A}$. This implies $x + y \in \tilde{V}$. We similarly have $\alpha x \in \tilde{V}$ for $\alpha \in F$ and $x \in \tilde{V}$. $\square$

1.3. Span of a set and Linear Independence.

Definition 3 (Linear combination of vectors). *Let V be a vector space over F and S be a non-empty subset of V . A vector $v \in V$ is called a linear combination of vectors in S (in short a linear combination of S) if there exist finite $v_1, \dots, v_n \in S$ and scalars $a_1, \dots, a_n \in F$ such that*

$$v = a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i.$$

We say that v is a linear combination of $v_1, \dots, v_n$ and call $a_1, \dots, a_n$ as the coefficients of the linear combination.

Example 3. (1) *Let $V = \mathbb{R}^2$. Then $(2, 4)$ is a linear combination of $(1, 0), (0, 1)$ since $2(1, 0) + 4(0, 1) = (2, 4)$. Similarly $(2, 4)$ is a linear combination of $(1, 1)$ and $(1, 0)$ as $(2, 4) = 4(1, 1) - 2(1, 0)$.*

(2) Let $V = M_{2 \times 2}(\mathbb{R})$. Then the identity

$$(1.1) \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

implies $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

Definition 4 (Span of a subset). Let S be a nonempty subset of a vector space V . The span of S is the set consisting of all finite linear combinations of the vectors in S , denoted by $\text{span}(S)$. We define $\text{span}(\phi) = \{0_V\}$.

Theorem 2 (span of a set is a subspace). The span of any subset S of a vector space V is a subspace of V . Furthermore, any subspace of V containing S must contain the span of S .

Example 4. (1) Let $S_1 = \{(1, 0), (0, -1)\}$. Then $\text{span}(S_1)$

$$= \{a(1, 0) + b(0, -1) : a, b \in \mathbb{R}\} = \{(a, -b) : a, b \in \mathbb{R}\} = \mathbb{R}^2.$$

(2) Let $S_2 = \{(1, 0, 1), (0, 0, 1)\}$, then $\text{span}(S_2) = \{(a, 0, a+b) : a, b \in \mathbb{R}\} = XZ$ plane.

(3) Let $S_3 = \left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$, then $\text{span}(S) =$ the set of all 2×2 real skew-symmetric matrices.

(4) $S_4 = \{v_1, \dots, v_n\}$ be a subset of a vector space V over F . Then

$$\text{span}(S_4) = \{a_1v_1 + \dots + a_nv_n : a_1, \dots, a_n \in F\}.$$

Definition 5 (Span for a vector space). A subset S of a vector space V generates V (spans V) if $\text{span}(S) = V$. In this case, we say that the vectors of S generate V (span V).

Proposition 1.2 (Spanning of $\mathbb{R}^3$). Let $S = \{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ be a subset of $\mathbb{R}^3$. Then

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \neq 0$$

if and only if $\text{span}(S) = \mathbb{R}^3$.

Definition 6 (Linear (in)dependence). Let S be a non-empty subset of a vector space V over F . We say S is linearly independent, if

$$a_1v_1 + \dots + a_nv_n = 0_V$$

implies $a_1 = \dots = a_n = 0$ for every $v_1, \dots, v_n \in S$. The set S is linearly dependent if there exists $a_1, \dots, a_n \in F$, $v_1, \dots, v_n \in S$, **not all** a_i **zero**, such that

$$a_1v_1 + \dots + a_nv_n = 0_V.$$

Example 5. (1) Let $S_1 = \{(1, 0), (0, -8)\}$, then S_1 is linearly independent.

(2) Let $S_2 = \{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ is linearly dependent as $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = (0, 0, 0)$.

(3) Let $S_3 = \{\sin x, \cos x\}$. Then S_3 is linearly independent as a subset of vector space V_4 given in Example 1.

(4) If S is a linearly independent subset of a vector space V , then any non-empty subset of S is also linearly independent.

Proposition 1.3. Let $S = \{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ be a subset of $\mathbb{R}^3$. Then

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \neq 0$$

if and only if S is linearly independent.

Definition 7 (Basis of a vector space). Let S be a subset of a vector space V over F . We say S to be a basis for V if

- $\text{span}(S) = V$,
- S is linearly independent in V .

Theorem 3 (Basis iff unique linear combination). Let $v_1, \dots, v_n$ be distinct vectors in a vector space V over F . Then $S = \{v_1, \dots, v_n\}$ is a basis for V if and only if every $v \in V$ can be written **uniquely** as a linear combinations of vectors of S , that is,

$$v = a_1v_1 + \dots + a_nv_n$$

for **unique** scalars $a_1, \dots, a_n$.

Proof. Let S be a basis for V . Since $\text{span}(S) = V$, for a given element $v \in V$, there is $a_1, \dots, a_n$ such that $v = a_1v_1 + \dots + a_nv_n$. To prove uniqueness, suppose

$$v = a_1v_1 + \dots + a_nv_n = b_1v_1 + \dots + b_nv_n.$$

This implies $(a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n = 0_V$. However, S is linearly independent, so $a_i = b_i$ for all i .

Conversely, suppose $v = a_1v_1 + \dots + a_nv_n$ for unique scalars $a_1, \dots, a_n$. Since $a_1v_1 + \dots + a_nv_n \in \text{span}(S)$, it follows that $\text{span}(S) = V$. Let $a_1v_1 + \dots + a_nv_n = 0_V = 0v_1 + \dots + 0v_n$. However by uniqueness of representation of v we have $a_i = 0$ for all i . $\square$

The above theorem gives an alternate definition of a basis, that is, S is a basis for V if every element of the vector space can be written uniquely as linear combination of vectors from S .

Example 6. (1) Let $S_1 = \{(1, 0), (0, 1)\}$. Then S_1 is a basis for $\mathbb{R}^2$. Let $i = 1, \dots, n$ and $e_i = (0, \dots, 1, \dots, 0)$ where 1 is at the i th place. Then $\{e_i \mid i = 1, \dots, n\}$ is a basis for $\mathbb{R}^n$, often called the standard basis for $\mathbb{R}^n$.

(2) Let $S_2 = \{(a_1, a_2, a_3), (b_1, b_2, b_3), (c_1, c_2, c_3)\} \subset \mathbb{R}^3$. By virtue of Proposition 1.2 and Proposition 1.3, S_2 is a basis if and only if

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \neq 0.$$

One can similarly generalise this result for $\mathbb{R}^n$.

(3) By equation (1.1) and using Theorem 3,

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is a basis for $M_{2 \times 2}(\mathbb{R})$.

Theorem 4 (Replacement Theorem). *Let V be a vector space over F and $S \subset V$ such that $|S| = n$ and $\text{span}(S) = V$. Let A be a linearly independent set in V with $|A| = m$. Then $m \leq n$ and there exists a subset S' of S with $|S'| = n - m$ such that $\text{span}(A \cup S') = V$.*

The following corollary provides some important applications of the replacement theorem.

Corollary 1.1. *Let V be a vector space of dimension n . Then the following holds.*

- (i) *Let S be a finite generating set for V containing at least n vectors. Then there exists a subset $S' \subset S$ such that $\text{span}(S') = V$, $|S'| = n$, and S' is a basis for V .*
- (ii) *Any linearly independent subset of V that contains exactly n vectors is a basis for V .*
- (iii) *Every linearly independent subset can be extended to a basis for V .*

Definition 8 (Dimension of a vector space). *Let S be a basis for a vector space V , and $|S|$ is finite. The cardinality of S , i.e. $|S|$, is independent of the choice of a basis and is called the **dimension** of the vector space V and denoted by $\dim(V)$.*

The fact that $|S|$ is independent of the choice of a basis requires proof. One may refer to Theorem 1.9, Theorem 1.10 (Theorem 4) and Corollary 1 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”. It is not necessary that $|S|$ is finite, hence if $|S|$ is not finite, we say V is an **infinite dimensional** vector space.

Example 7. (1) Let $V_1 = \mathbb{R}^n$, then by Example 6(1) $\dim(V_1) = n$.

(2) Let $V_2 = M_{m \times n}(\mathbb{R})$, then $\dim(V_2) = mn$.

(3) Let $V_3 = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is differentiable on } \mathbb{R}\}$. Then V_3 is infinite-dimensional as $\{x^n \mid n \in \mathbb{N}\}$ is a linearly independent set in V_3 .

Theorem 5 (Dimension of a subspace). *Let V be a finite-dimensional vector space over F and W be a subspace of V . Then W is finite dimensional and $\dim(W) \leq \dim(V)$. Furthermore, if $\dim(W) = \dim(V)$, then $W = V$.*

In the following examples, we discuss some typical applications of propositions, theorems and corollaries discussed in this chapter.

Example 1: Prove that $\text{span}(\{(1, 2), (3, 7)\}) = \mathbb{R}^2$.

Solution: Since $\{(1, 2), (3, 7)\}$ is linearly independent, and $|\{(1, 2), (3, 7)\}| = 2$, using Corollary 1.1, we see that $\{(1, 2), (3, 7)\}$ is a basis for $\mathbb{R}^2$.

Example 2: Let W be the space of 2×2 symmetric matrices with real entries. Find $\dim(W)$.

Solution: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W$, then $A = A^t$, where A^t denotes the transpose of the matrix

A . This implies A must be of the form $\begin{bmatrix} a & b \\ b & d \end{bmatrix}$. Note that

$$\begin{bmatrix} a & b \\ b & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

i.e. A can be written uniquely as a linear combination of vectors from $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

Using Theorem 3, $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for W and $\dim(W) = 3$.

Example 3: Let W_1, W_2 be two subspaces of V , then prove that

$$W_1 + W_2 := \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\},$$

is a subspace of V .

Solution: Since $0_V = 0_V + 0_V$ and $0_V \in W_1 \cap W_2$, it follows that $0_V \in W_1 + W_2$. Let $u, v \in W$, then $u = u_1 + u_2, v = v_1 + v_2$, where $u_1, v_1 \in W_1$ and $u_2, v_2 \in W_2$. By (VS1) and (VS2) (see Definition 1), $u + v = (u_1 + v_1) + (u_2 + v_2) \in W_1 + W_2$ as $u_1 + v_1 \in W_1$ and $u_2 + v_2 \in W_2$. Let $w = w_1 + w_2 \in W$, we have $aw = aw_1 + aw_2$, where $a \in F$. Since $aw_1 \in W_1$ and $aw_2 \in W_2$, the proof is complete by Theorem 1.

2. LINEAR MAPS

In this section, we consider maps between vector spaces that preserve their structure.

2.1. Range space and null space.

Definition 9 (Linear map(transformation)). *Let V and W be two vector spaces over a field F . A function T from V to W is said to be a linear map (linear transformation) if the following are satisfied.*

- (i) $T(x + y) = T(x) + T(y)$ for all $x, y \in V$.
- (ii) $T(ax) = aT(x)$ for all $a \in F$ and $x \in V$.

Example 8. (1) Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T_1(a_1, a_2) = (0, a_2)$. T_1 can be seen as the projection on the Y -axis.

(2) Let $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T_2(a_1, a_2) = (a_2, a_1)$. T can be seen as the reflection about the line $y = x$ in $\mathbb{R}^2$.

(3) Let $T_3 : M_{3 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}$ be given by $T_3(A) = \text{Trace}(A)$, where $\text{Trace}(A)$ denotes the trace of the matrix A .

A linear map T for which $V = W$ is called a linear operator.

Proposition 2.1 (0_V maps to 0_W). *Let V, W be two vector spaces over F with zero vectors 0_V and 0_W respectively. Let $T : V \rightarrow W$ be a linear map. Then $T(0_V) = 0_W$.*

Proof. We have $0_V + 0_V = 0_V$. Applying T on both sides and using the linearity of T , one obtains $T(0_V) + T(0_V) = T(0_V)$. The result then follows from cancelation of $T(0_V)$. $\square$

Proposition 2.2 ($\text{Im}(T)$ is a subspace of W). *Let V, W be two vector spaces over F , and $T : V \rightarrow W$ be a linear map. The set $\text{Im}(T) := \{T(v) : v \in V\}$ is a subspace of W .*

Proof. By Proposition 2.1, it follows that $0_W \in \text{Im}(T)$. Let $w_1, w_2 \in \text{Im}(T)$, then by definition of $\text{Im}(T)$, there is $v_1, v_2 \in V$ such that $T(v_1) = w_1, T(v_2) = w_2$. However, T being a linear map implies,

$$w_1 + w_2 = T(v_1) + T(v_2) = T(v_1 + v_2) \in \text{Im}(T).$$

Let $a \in F, w \in \text{Im}(T)$. We can write $w = T(v)$ for some $v \in V$. Note that

$$\alpha w = \alpha T(v) = T(\alpha v) \in \text{Im}(T).$$

Hence $\text{Im}(T)$ is a subspace of W by Theorem 1. $\square$

Definition 10 (Rank of a linear map). *Let V, W be two vector spaces over F , and let $T : V \rightarrow W$ be a linear map. The set $\text{Im}(T) = \{T(v) | v \in V\}$ is called the range space (Image space) of T . The dimension of $\text{Im}(T)$ is called the **rank** of T and is denoted by $\text{rank}(T)$.*

For a linear map $T : V \rightarrow W$, $\text{Im}(T)$ is a natural subspace of W . The vector space V also has a natural subspace given as follows. Consider $\ker(T) = \{v \in V | T(v) = 0_W\}$. Using Proposition 2.1, one easily checks that $\ker(T)$ is a subspace of V . The space $\ker(T)$ is called the Kernel space or Null space of the linear map T .

Definition 11 (Nullity of a linear map). *The dimension of $\ker(T)$ is called the **nullity** of the linear map T and is denoted by $\text{nullity}(T)$.*

Proposition 2.3. *Let T be a linear map from a vector space V to the vector space W . Let $B = \{v_1, \dots, v_n\}$ be a basis for V , then $\text{Im}(T) = \text{span}(\{T(v_1), \dots, T(v_n)\})$.*

Proof. Clearly $T(v_i) \in \text{Im}(T)$ for all i . $\text{Im}(T)$ being a subspace contains all finite linear combinations of elements from $\{T(v_1), \dots, T(v_n)\}$, which implies $\text{span}(\{T(v_1), \dots, T(v_n)\}) \subset \text{Im}(T)$. Conversely, if $w \in \text{Im}(T)$, we can write $w = T(v)$ for some $v \in V$. There is $a_1, \dots, a_n \in F$ such that $v = a_1v_1 + \dots + a_nv_n$ as B is a basis for V . However T being linear implies

$$w = T(a_1v_1 + \dots + a_nv_n) = a_1T(v_1) + \dots + a_nT(v_n) \in \text{span}(\{T(v_1), \dots, T(v_n)\}).$$

□

2.2. Rank Nullity Theorem. We will prove the rank-nullity theorem for a linear map defined on a finite-dimensional space and see some examples for verification.

Theorem 6 (Rank Nullity Theorem). *Let V be a finite-dimensional vector space. Let T be a linear map from V to a vector space W . Then*

$$\dim(V) = \text{nullity}(T) + \text{rank}(T).$$

Proof. Let $\{v_1, \dots, v_k\}$ be a basis for $\ker(T)$. Using corollary to Theorem 1.11 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition” (alternatively, Corollary 1.1(iii)), we can extend $\{v_1, \dots, v_k\}$ to a basis $B = \{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ of V . Using Proposition 2.3,

$$\text{Im}(T) = \text{span}(\{T(v_1), \dots, T(v_k), T(v_{k+1}), \dots, T(v_n)\}) = \text{span}(\{T(v_{k+1}), \dots, T(v_n)\})$$

as $T(v_i) = 0_W$ for $i = 1, \dots, k$.

We are left to show that $\{T(v_{k+1}), \dots, T(v_n)\}$ is linearly independent. Let $a_1T(v_{k+1}) + \dots + a_{n-k}T(v_n) = 0_W$ for some scalars $a_i \in F$. Using linearity of T , we have $a_1v_{k+1} + \dots + a_{n-k}v_n \in \ker(T)$. However $\{v_1, \dots, v_k\}$ being a basis for $\ker(T)$, for some scalars b_i for $i = 1, \dots, k$, we get

$$\begin{aligned} a_1v_{k+1} + \dots + a_{n-k}v_n &= b_1v_1 + \dots + b_kv_k \\ \implies b_1v_1 + \dots + b_kv_k + (-a_1)v_{k+1} + \dots + (-a_{n-k})v_n &= 0_V. \end{aligned}$$

But B is a basis for V , hence is linearly independent implying $a_i = 0$, $b_j = 0$ for all i, j . We have shown $\{T(v_{k+1}), \dots, T(v_n)\}$ is a basis for $\text{Im}(T)$. By definition, $\text{nullity}(T) = |\{v_1, \dots, v_k\}| = k$ and $\text{rank}(T) = |\{T(v_{k+1}), \dots, T(v_n)\}| = n - k$. This completes the proof. □

2.2.1. Verification of rank nullity Theorem with examples. In this subsection, we verify the rank-nullity theorem via four examples.

Example 1: Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x, y)$. The kernel is all vectors that get mapped to $(0, 0)$. We have

$$T(x, y, z) = (0, 0) \iff x = 0, y = 0, z \in \mathbb{R}.$$

So

$$\ker(T) = \{(0, 0, z) : z \in \mathbb{R}\} = \text{span}(\{(0, 0, 1)\}),$$

which has dimension 1.

The image is all of $\mathbb{R}^2$, so $\dim(\text{Im}(T)) = 2$.

Verification:

$$\dim(\mathbb{R}^3) = \text{rank}(T) + \text{nullity}(T) = 2 + 1 = 3.$$

Example 2: Linear Map on $\mathbb{R}^2$ Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) = (x + y, x + 2y)$.

On solving $x + y = 0$ and $x + 2y = 0$, we get $x = y = 0$. So the kernel is $\{(0, 0)\}$, which is of dimension 0.

The vectors $(1, 1)$ and $(1, 2)$ are linearly independent; therefore, the image is $\mathbb{R}^2$ (Why?), which is of dimension 2.

Verification:

$$\dim(\mathbb{R}^2) = \text{rank}(T) + \text{nullity}(T) = 2 + 0 = 2.$$

Example 3: Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2 + x_3, x_3 + x_4)$.

On setting $T(x_1, x_2, x_3, x_4) = (0, 0, 0)$, we have

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 + x_3 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

After solving, one obtains

$$x_1 = -x_2, \quad x_3 = -x_2, \quad x_4 = x_2.$$

Hence $\ker(T) = \{(-a, a, -a, a) \mid a \in \mathbb{R}\} = \text{span}(\{(-1, 1, -1, 1)\})$ dimension 1. Note that

$$T(1, 0, 0, 0) = (1, 0, 0), T(0, 1, 0, 0) = (1, 1, 0), T(0, 0, 1, 0) = (0, 1, 1), T(0, 0, 0, 1) = (0, 0, 1).$$

Using Proposition 2.3 and Proposition 1.2,

$$\text{Im}(T) = \text{span}(\{(1, 0, 0), (1, 1, 0), (0, 1, 1), (0, 0, 1)\}) = \text{span}(\{(1, 0, 0), (1, 1, 0), (0, 0, 1)\}) = \mathbb{R}^3.$$

Verification:

$$\dim(\mathbb{R}^4) = \text{rank}(T) + \text{nullity}(T) = 3 + 1 = 4.$$

Example 4: Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}$ be $T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = a + d$.

T sends matrices with zero trace to 0, which implies

$$\ker(T) = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} : a, b, c \in \mathbb{R} \right\}.$$

Observe that $\begin{bmatrix} a & b \\ c & -a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, which implies any vector of $\ker(T)$

can be written uniquely as linear combination of vectors $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$. Hence by Theorem 3,

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

is a basis for $\ker(T)$. Trace of $\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$ is a , for any real number a , so the image is $\mathbb{R}$, which is of dimension 1.

Verification:

$$\dim(M_{2 \times 2}(\mathbb{R})) = \text{rank}(T) + \text{nullity}(T) = 1 + 3 = 4.$$

2.3. Representation of a linear map by matrices. Consider a matrix $A \in M_{m \times n}(\mathbb{R})$, where $A = (a_{ij})$. We begin this section with a definition of an ordered basis.

Definition 12. Let V be a finite dimensional vector space. An ordered basis for V is a basis for V with a specific order.

As an example, consider $S_1 = \{(1, 0), (0, 1)\}$ and $S_2 = \{(0, 1), (1, 0)\}$. Both S_1 and S_2 are ordered bases, but $S_1 \neq S_2$ as ordered bases are not equal.

Let V and W be two finite dimensional vector spaces. Consider a linear map $T : V \rightarrow W$. Let $B_V = \{v_1, v_2, \dots, v_n\}$ and $B_W = \{w_1, w_2, \dots, w_m\}$ be ordered bases for V and W respectively. For a given j , consider the vector $v_j \in B_V$. By Theorem 3, $T(v_j)$ can be uniquely written as a linear combination of basis vectors for W , i.e.,

$$T(v_j) = \sum_{i=1}^m a_{ij} w_i$$

for some scalars $a_{ij} \in \mathbb{R}$. For a fixed j , we call (a_{ij}) as the coordinates of $T(v_j)$ w.r.t. basis B_W . Note that (a_{ij}) is a collection of m scalars which can be seen as a column matrix in the usual sense. Similarly, on writing the coordinates of $T(v_j)$ for all $1 \leq j \leq n$ together in increasing order, we get a matrix of order $m \times n$ as there are n column matrices, each matrix containing m no of scalars. We denote this obtained matrix by $[T]_{B_V}^{B_W}$ whose j th column is the coordinates of $T(v_j)$ w.r.t. the ordered basis B_W . If $V = W$ and $B_V = B_W$, then, we write $[T]_{B_V}$ instead of $[T]_{B_V}^{B_W}$. For example, refer to Examples 3 and 4 of Section 2.2 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”.

Conversely, given $A = (a_{ij})_{i=1:m, j=1:n} \in M_{m \times n}(\mathbb{R})$, we define $L_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by

$$L_A(x_1, \dots, x_n) = \left(\sum_{j=1}^n a_{1j} x_j, \sum_{j=1}^n a_{2j} x_j, \dots, \sum_{j=1}^n a_{mj} x_j \right).$$

One easily checks that L_A is a linear map, which is called a left multiplication transformation.

Example 9. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 7 \end{bmatrix} \in M_{2 \times 3}(\mathbb{R})$. The left multiplication transformation is given by $L_A(x, y, z) = (x + 2y + 3z, -y + 7z)$.

Theorem 7. Let $A \in M_{m \times n}(\mathbb{R})$ and $L_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be its corresponding left-multiplication transformation. Let $B_{\mathbb{R}^n}$ and $B_{\mathbb{R}^m}$ be standard bases (see Example 6(1)) for $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively. Then the following holds.

- (1) $[L_A]_{B_{\mathbb{R}^m}}^{B_{\mathbb{R}^n}} = A$.
- (2) $L_A = L_C$ if and only if $A = C$.
- (3) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, then there is a unique $C \in M_{m \times n}$ such that $T = L_C$. Furthermore, $C = [T]_{B_{\mathbb{R}^m}}^{B_{\mathbb{R}^n}}$.
- (4) If $E \in M_{n \times p}(\mathbb{R})$, then $L_{AE} = L_A \circ L_E$.

3. DIAGONALISATION

3.1. Eigenvalues and Eigenvectors. We start with some definitions.

Definition 13. Let V be a finite-dimensional vector space. A linear operator on V is said to be diagonalisable if there is an ordered basis B of V such that $[T]_B$ is a diagonal matrix. A square matrix A is diagonalisable if L_A is diagonalisable.

Definition 14 (Eigenvalues). Let T be a linear operator on a finite-dimensional vector space V . A nonzero vector v is called an eigenvector of T if there exists $\lambda \in \mathbb{R}$ such that $T(v) = \lambda v$. The scalar λ is called the eigenvalue corresponding eigenvector v .

Let $A \in M_{n \times n}(\mathbb{R})$. A nonzero vector $x \in M_{n \times 1}(\mathbb{R})$ is called an eigenvector if x^t is an eigen vector for L_A , i.e., $L_A(x^t) = \lambda x^t$ for some $\lambda \in \mathbb{R}$. The scalar λ is called the eigenvalue of A corresponding to the eigenvector v .

Theorem 8. Let $A \in M_{n \times n}(\mathbb{R})$. Then $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Definition 15. Let $A \in M_{n \times n}(\mathbb{R})$. The polynomial $f(x) = \det(A - xI_n)$ is called the characteristic polynomial of A .

Let T be a linear operator in a finite-dimensional vector space V . Let B be an ordered basis for V . The determinant and the characteristic polynomial of T is given to be $\det([T]_B)$ and $\det([T]_B - xI_n)$ respectively.

For example, refer to Examples 3 to 7 of Section 5.1 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”.

Theorem 9 (Cayley-Hamilton Theorem). Let T be a linear operator in a finite-dimensional vector space V or $A \in M_{\dim(V) \times \dim(V)}(\mathbb{R})$ with characteristic polynomial $f(t)$. Then $f(T) = O$ (O denotes the zero map on V) or $f(A) = O_{n \times n}$ respectively.

Theorem 10. Let $A \in M_{n \times n}(\mathbb{R})$ and $\lambda \in \mathbb{R}$. A vector $v \in V$ is an eigenvector of A corresponding to λ if and only if $v \neq 0$ and $(A - \lambda I)v = 0$.

3.1.1. Properties of eigenvalues. In this subsection, we discuss some properties concerning eigenvalues of a matrix $A \in M_{n \times n}(\mathbb{R})$. It is recommended to see a proof of these facts.

- (1) If λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k for $k \in \mathbb{N}$.
- (2) If A has multiplicative inverse with an eigenvalue λ , then λ^{-1} is an eigenvalue of A^{-1} .
- (3) Let $\lambda_1, \dots, \lambda_n$ be all eigenvalues of A . Then $\text{trace}(A) = \sum_{i=1}^n \lambda_i$ and $\det(A) = \prod_{i=1}^n \lambda_i$.
- (4) Let $P(x)$ be a real polynomial. If λ is an eigenvalue of A , then $P(\lambda)$ is an eigenvalue of $P(A)$.
- (5) A symmetric matrix always has real eigenvalues.
- (6) Two matrices A and B are similar if $A = PBP^{-1}$ for some invertible matrix P . Both A and B have the same set of eigenvalues.

3.2. Diagonalisibility.

Definition 16 (Algebraic multiplicity). Let λ be an eigenvalue of a linear operator or matrix with the characteristic polynomial $f(x)$. The algebraic multiplicity of λ is the largest positive integer k such that $(x - \lambda)^k$ divides $f(x)$.

Definition 17 (Eigenspace). Let T be a linear operator on the vector space V and λ be an eigen value of T . The subspace $E_\lambda = \{v \in V : T(v) = \lambda v\} = N(T - \lambda I_V)$ is called the eigenspace of T for the eigenvalue λ . Similarly, the eigenspace of a square matrix A for an eigenvalue λ is defined to be the eigenspace of L_A corresponding to λ .

Definition 18. (Geometric multiplicity) Let λ be an eigenvalue of a linear operator T on a finite-dimensional vector space V . The geometric multiplicity of λ is equal $\dim(E_\lambda) = \dim(N(T - \lambda I_V)) = \dim(V) - \text{rank}(T - \lambda I_V)$.

Lemma 3.1. Let λ be an eigenvalue of a linear operator T on a finite-dimensional vector space V or $A \in M_{\dim(V) \times \dim(V)}(\mathbb{R})$. Then the algebraic multiplicity of λ is always greater than or equal to the geometric multiplicity of λ .

Theorem 11 (Criterion for Diagonalisiblity). Let $A \in M_{n \times n}(\mathbb{R})$ be such that all its eigenvalues are real numbers. Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of A . Then A is diagonalizable if and only if the algebraic multiplicity of λ_i is equal to the geometric multiplicity of λ_i for all $i = 1, \dots, k$.

Let $A \in M_{n \times n}(\mathbb{R})$ be a diagonalizable matrix. Then we can write $A = PDP^{-1}$, where P has as its columns the vectors in a basis of eigenvectors of A , and D has its j th diagonal entry the eigenvalue of A corresponding to the j th column of P . Diagonalisation helps us to compute the powers of A , since $A^m = PD^mP^{-1}$. For example, refer to Examples 2 to 7 of Section 5.2 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”.

4. INNER PRODUCT SPACES

Definition 19. Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$. An inner product on V is a function that assigns an element in $\mathbb{R}$ or $\mathbb{C}$ to every ordered pair of $x, y \in V$, denoted by $\langle x, y \rangle$, such that the following hold:

- (1) $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$ for all $x, y, z \in V$.
- (2) $\langle cx, y \rangle = c\langle x, y \rangle$ for $c \in \mathbb{R}$ or $c \in \mathbb{C}$.
- (3) $\overline{\langle x, y \rangle} = \langle y, x \rangle$ where bar denotes the complex conjugation.
- (4) If $x \neq 0$, then $\langle x, x \rangle \in \mathbb{R}$ and $\langle x, x \rangle > 0$.

For example, refer to Examples 1 to 5 of Section 6.1 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”. In the following theorem, we discuss some results in an inner product space.

Theorem 12. Let V be an inner product space over $\mathbb{C}$. Let $x, y, z \in V$ and $c \in \mathbb{R}$. The following statements are true.

- (a) $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$.
- (b) $\langle x, cy \rangle = \bar{c}\langle x, y \rangle$.
- (c) $\langle x, 0 \rangle = \langle 0, x \rangle = 0$.
- (d) $\langle x, x \rangle = 0$ if and only if $x = 0$.
- (e) If $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V$, then $y = z$.

Definition 20. Let V be an inner product space. A set $S \subset V$ is called an orthogonal set if $\langle v_i, v_j \rangle = 0$ for every $i \neq j$ with $v_i, v_j \in S$.

An orthogonal set is called orthonormal if $\langle v, v \rangle = 1$ for all $v \in S$.

Consider a basis $v_1, \dots, v_n$ of an inner product space V which is an orthogonal set. Using Theorem 3, every $v \in V$ can be written as

$$v = \sum_{i=1}^n \alpha_i v_i,$$

for some scalars α_i . Considering $\langle v_i, v \rangle$, it can be easily shown that $\alpha_i = \frac{\langle v_i, v \rangle}{\langle v_i, v_i \rangle}$ for all $i = 1, \dots, n$. Furthermore, if the basis is an orthonormal set, then $\alpha_i = \langle v_i, v \rangle$. Hence, making an orthonormal set out of a basis becomes useful as the coefficients α_i can be easily computed by taking the inner product of a given vector v with one of the basis vectors v_i . In the next subsection, we discuss on the process of getting an orthonormal basis out of any basis using Gram-Schmidt Orthogonalisation. We end this section with the norm of a vector and some of its properties.

Definition 21. Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$. The norm or length of $x \in V$ is defined as $\sqrt{\langle x, x \rangle}$ and is denoted by $\|x\|$.

The following properties of the norm are immediate.

Theorem 13. Let V be an inner product space over $\mathbb{C}$. Let $x, y, z \in V$ and $c \in \mathbb{R}$. The following statements are true.

- (a) $\|cx\| = |c| \times \|x\|$.
- (b) $\|x\| \geq 0$. $\|x\| = 0$ if and only if $x = 0$.
- (c) (Cauchy-Schwarz Inequality) $|\langle x, y \rangle| \leq \|x\| \|y\|$.

- (d) (Triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.
(e) If $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V$, then $y = z$.

4.1. Gram Schmidt Orthogonalisation. Let V be a finite-dimensional inner product space with a linearly independent set $\{w_1, \dots, w_n\}$. Gram-Schmidt Orthogonalisation process gives us a step-by-step procedure to obtain an orthogonal set out of this linearly independent set without changing $\text{span}(\{w_1, \dots, w_n\})$.

Theorem 14. Let V be an inner product space and $S = \{w_1, \dots, w_n\}$ be a linearly independent subset of V . Let $S' = \{v_1, \dots, v_n\}$, where $v_1 = w_1$ and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j \quad \text{for } 2 \leq k \leq n.$$

Then S' is an orthogonal set of nonzero vectors with $\text{span}(S') = \text{span}(S)$.

If $S'' = \left\{ \frac{v_1}{\|v_1\|}, \dots, \frac{v_n}{\|v_n\|} \right\}$, then S'' is an orthonormal set of nonzero vectors with $\text{span}(S'') = \text{span}(S)$.

For example, refer to Examples 4 to 6 of Section 6.2 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”. Some other relevant webpages are Link 1 and Link 2.

Theorem 15. Let V be a nonzero finite-dimensional inner product space. Then V has an orthonormal basis $B = \{v_1, \dots, v_n\}$. Furthermore, if $x \in V$, then

$$x = \sum_{i=1}^n \langle x, v_i \rangle v_i.$$

For example, refer to Examples 7 to 10 of Section 6.2 of the book “S. H. Friedberg, A. J. Insel and L. E. Spence, Linear Algebra, Fourth Edition”.

DR. JISHU DAS, DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES (SECTION: MATHEMATICS),
COEP TECHNOLOGICAL UNIVERSITY, PUNE.

Email address: dasj.maths@coeptech.ac.in